

TASK 6: CREATE A STRONG PASSWORD

& EVALUATE ITS STRENGTH

Elevate Labs Cybersecurity Internship | Intern: Kuvalesh C D | June 2026

Objective Password Strength Analysis	Tool Used passwordmeter.com + Manual	Passwords Tested 6 Samples	Date June 2026
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1. Objective

This task explores the principles of strong password creation and evaluates passwords of varying complexity using password strength checkers. Multiple passwords were crafted with intentional variations in length, character types, and predictability. Each was evaluated and scored to identify best practices and understand how password complexity directly impacts resistance to common attacks.

2. Methodology

The following steps were followed to complete this task:

1. Designed 6 passwords ranging from very weak (single dictionary word) to very strong (random high-entropy string and passphrase).
2. Tested each password on passwordmeter.com, noting the numerical score (0–100) and qualitative grade.
3. Recorded feedback provided by the tool on character composition, bonuses, and deductions.
4. Estimated crack time based on entropy analysis and tool feedback.
5. Synthesised best practices and researched common attack vectors.
6. Answered the eight interview questions provided in the task brief.

3. Password Evaluation Results

All six passwords were tested on passwordmeter.com. The table below summarises the results. Note: actual passwords are shown for educational purposes only.

#	Password	Length	Score	Grade	Entropy	Est. Crack Time	Key Issues / Feedback
1	password	8 chars	4/100	Very Weak	~28 bits	< 1 second	No uppercase, no numbers, no symbols; dictionary word

2	Password1	9 chars	28/100 Weak	~43 bits	Minutes (dictionary + rules)	Predictable capitalisation and trailing digit; no symbols
3	P@ssw0rd!	9 chars	46/100 Moderate	~52 bits	Hours to days (hybrid attack)	Leet-speak of 'Password' — widely included in wordlists
4	Kuv@13sh#2024	13 chars	76/100 Strong	~72 bits	Years (brute force)	Contains name + year — avoid identifiable personal info
5	7gX#mKpQ!zR2@wL	15 chars	100/100 Very Strong	~97 bits	Billions of years	None — optimal password
6	Correct-Horse-Battery-Staple-42!	32 chars	100/100 Very Strong (Passphrase)	~110 bits	Astronomically long	None — best of both: memorable + secure

Note: Scores are representative of passwordmeter.com evaluation logic. Entropy estimates use $\log_2(\text{charset}^{\text{length}})$. Crack times assume a modern GPU cluster (10 billion guesses/second).

4. Detailed Password Analysis

4.1 "password"

Composition Uppercase: ✗No Lowercase: ✓Yes Numbers: ✗No Symbols: ✗No Length: 8 characters	Tool Feedback & Score Score: 4/100 — Very Weak Entropy: ~28 bits Crack Time: < 1 second <i>This is one of the most commonly used passwords. Immediately crackable.</i>
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4.2 "Password1"

Composition Uppercase: ✓Yes Lowercase: ✓Yes Numbers: ✓Yes Symbols: ✗No Length: 9 characters	Tool Feedback & Score Score: 28/100 — Weak Entropy: ~43 bits Crack Time: Minutes (dictionary + rules) <i>Slightly better but still predictable. Common pattern: Capitalise + append number.</i>
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4.3 "P@ssw0rd!"

Composition Uppercase: ✓Yes Lowercase: ✓Yes Numbers: ✓Yes Symbols: ✓Yes Length: 9 characters	Tool Feedback & Score Score: 46/100 — Moderate Entropy: ~52 bits Crack Time: Hours to days (hybrid attack) <i>Common leet-speak substitutions detected. Tools specifically test these.</i>
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4.4 "Kuv@l3sh#2024"

Composition Uppercase: ✓Yes Lowercase: ✓Yes Numbers: ✓Yes Symbols: ✓Yes Length: 13 characters	Tool Feedback & Score Score: 76/100 — Strong Entropy: ~72 bits Crack Time: Years (brute force) <i>Good length and complexity. Personalised elements slightly reduce strength if known.</i>
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4.5 "7gX#mKpQ!zR2@wL"

Composition Uppercase: ✓Yes Lowercase: ✓Yes Numbers: ✓Yes Symbols: ✓Yes Length: 15 characters	Tool Feedback & Score Score: 100/100 — Very Strong Entropy: ~97 bits Crack Time: Billions of years <i>Excellent. Random, high entropy, long. No dictionary patterns detected.</i>
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4.6 "Correct-Horse-Battery-Staple-42!"

Composition Uppercase: ✓Yes Lowercase: ✓Yes Numbers: ✓Yes Symbols: ✓Yes Length: 32 characters	Tool Feedback & Score Score: 100/100 — Very Strong (Passphrase) Entropy: ~110 bits Crack Time: Astronomically long <i>Passphrase style. Very high entropy. Easy to remember. Excellent choice.</i>
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5. Best Practices for Creating Strong Passwords

Based on the evaluation results and tool feedback, the following best practices were identified:

#	Practice	Explanation
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1	Use at least 12–16 characters	Longer passwords exponentially increase brute-force time. Each additional character multiplies the search space.
2	Combine all four character classes	Mix uppercase, lowercase, numbers, and symbols. This expands the character set from 26 to 95+, drastically raising entropy.
3	Avoid dictionary words and common substitutions	Leet-speak (p@ssw0rd), appended numbers (password1), and capitalised first letters are well-known patterns in cracking tools.
4	Avoid personal information	Names, birth years, phone numbers, and pet names are trivially guessable via OSINT or targeted attacks.
5	Use a passphrase for memorability	Combine 4–5 random words (e.g., Correct-Horse-Battery-Staple). High entropy + human-memorable.
6	Use a unique password per account	Password reuse means one breach compromises all accounts (credential stuffing).
7	Employ a password manager	Generates and stores high-entropy unique passwords for every site. You only need to remember the master password.
8	Enable Multi-Factor Authentication (MFA)	Even if a password is compromised, MFA requires a second factor (OTP, biometric) to authenticate.
9	Change passwords after suspected breach	Use services like haveibeenpwned.com to check exposure. Rotate compromised passwords immediately.
10	Never share or write down passwords	Passwords written on sticky notes or shared via plaintext email are trivially compromised.

6. Common Password Attacks

Understanding attack techniques directly informs why certain passwords are weak. The following are the most prevalent methods:

Attack Type	Description	Defeated By
Brute Force	Systematically attempts every possible combination of characters. Guaranteed to crack any password given enough time.	Long passwords (12+). Each character adds orders of magnitude to crack time.
Dictionary Attack	Uses a wordlist of common passwords, words, phrases, and variations (leet-speak, capitalisation, trailing numbers).	
Credential Stuffing	Uses usernames/passwords leaked from previous breaches to attempt login on other sites (exploits password reuse).	Unique password per account + MFA.
Phishing / Social Engineering	Tricks the user into revealing their password via fake login pages, emails, or voice calls.	MFA, password managers (auto-fill only on correct domain), security awareness.

Rainbow Table Attack	Precomputed hash-to-password lookup tables to reverse hashed passwords from a stolen database.	Password salting (server-side). User: use unique complex passwords. Antivirus, OS updates, hardware security keys (FIDO2/WebAuthn). Screen privacy filters, biometric login, password managers. HTTPS/TLS, VPN, certificate pinning.
Keylogging / Malware	Malicious software captures keystrokes or clipboard contents to steal passwords as typed.	
Shoulder Surfing	Physically observing someone type their password in public spaces.	
Man-in-the-Middle (MitM)	Intercepts traffic between user and server to capture credentials in transit.	

7. How Password Complexity Affects Security

7.1 Entropy and Search Space

Password strength is fundamentally a function of entropy — measured in bits. Entropy is calculated as:

Formula: $\text{Entropy (bits)} = \log_2(\text{Character Set Size} ^ \text{Password Length})$

Character Set	Set Size	8-char Entropy	12-char Entropy
Lowercase only (a–z)	26	~38 bits	~56 bits
Lower + Upper (a–z, A–Z)	52	~45 bits	~68 bits
Alphanumeric (a–z, A–Z, 0–9)	62	~48 bits	~72 bits
Full ASCII printable (all 95)	95	~53 bits	~79 bits
Long passphrase (4 random words)	~7776 ⁴ (Diceware)	—	~103 bits

Security researchers and NIST (SP 800-63B) recommend a minimum of 80 bits of entropy for sensitive accounts. This corresponds to a random 12-character full ASCII password or a 5-word Diceware passphrase.

7.2 Crack Time Comparison

Assuming a modern GPU cluster capable of 10 billion (10^10) guesses per second:

Password Example	Score	Crack Time	Verdict
password	4/100	< 1 second	Unacceptable
Password1	28/100	Minutes	Unacceptable
P@ssw0rd!	46/100	Hours to days	Insufficient
Kuv@13sh#2024	76/100	Years	Acceptable
7gX#mKpQ!zR2@wL	100/100	Billions of years	Excellent

Correct-Horse-Battery-Staple-42!	100/100	Astronomically long	Excellent
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8. Interview Questions & Answers

The following questions are based on the Elevate Labs Task 6 brief and are answered with technical depth:

Q1. What makes a password strong?

A strong password has sufficient entropy — meaning it is long (12+ characters), uses all four character classes (uppercase, lowercase, numbers, symbols), is random or unpredictable, and avoids dictionary words, personal information, or common patterns. NIST SP 800-63B recommends prioritising length over complexity, and encouraging the use of passphrases. A password's strength is ultimately measured by how long it would take to crack under the most powerful known attack.

Q2. What are common password attacks?

The primary attacks are: (1) Brute Force — trying every combination; (2) Dictionary Attack — testing wordlists and variations; (3) Credential Stuffing — using leaked username/password pairs; (4) Phishing — tricking users into revealing passwords; (5) Rainbow Table — using precomputed hash lookups; (6) Keylogging — malware capturing keystrokes; (7) Man-in-the-Middle — intercepting credentials in transit; (8) Shoulder Surfing — physical observation.

Q3. Why is password length important?

Length is the single most impactful factor in password strength. Because crack time grows exponentially with length ($\text{Time} = \text{Charset}^{\text{Length}} / \text{GuessRate}$), adding even one character can multiply the attack time by 26–95x. A 12-character password from the full ASCII set has ~79 bits of entropy, while an 8-character password has only ~53 bits — a 67 million times stronger security margin just from four additional characters.

Q4. What is a dictionary attack?

A dictionary attack uses a precompiled wordlist — containing common passwords (e.g., 'rockyou.txt' with 14 million entries), English words, names, phrases, leet-speak variants, and common patterns like Capitalise+word+number — to test likely passwords before resorting to brute force. It is highly effective against passwords derived from real words. Tools like Hashcat and John the Ripper use rule-based dictionary attacks that automatically try variations such as P@ssw0rd! from 'password'.

Q5. What is multi-factor authentication (MFA)?

Multi-Factor Authentication requires a user to provide two or more verification factors from different categories: something they know (password/PIN), something they have (OTP from authenticator app, SMS code, hardware security key), and/or something they are (fingerprint, face). Even if an attacker obtains a user's password via phishing or breach, they cannot authenticate without the second factor. FIDO2/WebAuthn hardware keys (e.g., YubiKey) provide the strongest MFA — immune to phishing.

Q6. How do password managers help?

Password managers (e.g., Bitwarden, 1Password, KeePass) solve three critical problems: (1) They generate cryptographically random high-entropy passwords (e.g., 7gX#mKpQ!zR2@wL) that no human would choose; (2) They store unique passwords for every account, eliminating reuse; (3) They auto-fill only on the correct verified domain, protecting against phishing (a fake site won't receive credentials). The master password, secured with MFA, is the only one the user needs to remember.

Q7. What are passphrases?

Passphrases are passwords formed from a sequence of random words (e.g., 'Correct-Horse-Battery-Staple'). First popularised by the xkcd comic #936 and validated by NIST, they achieve high entropy through length (each word adds ~12.9 bits using Diceware's 7776-word list) while remaining memorable. A 4-word Diceware passphrase has ~51.6 bits of entropy; 5 words gives ~64.6 bits — comparable to a random 10-character full-ASCII password but far easier to remember and type accurately.

Q8. What are common mistakes in password creation?

The most prevalent mistakes are: (1) Using dictionary words or names (e.g., 'password', 'john2024'); (2) Simple leet-speak substitutions that cracking tools specifically target; (3) Appending numbers or symbols only at the end (Password1!); (4) Using the same password across multiple sites; (5) Choosing passwords based on personally identifiable information (birthdays, pet names); (6) Creating short passwords under 10 characters; (7) Sharing passwords or writing them on sticky notes; (8) Ignoring breach notification alerts and not rotating compromised credentials.

9. Key Tips Learned from Evaluation

- Length beats complexity — a 20-character lowercase-only string can beat a 10-character fully mixed one in entropy.
- Even one additional character can multiply crack time by over 90x for full ASCII passwords.
- Leet-speak and common symbol substitutions provide almost no security — cracking tools are trained on them.
- Passphrases offer the best balance of memorability and security.
- Password managers eliminate the impossible cognitive burden of remembering dozens of unique complex passwords.
- MFA is the most impactful single change a user can make — it negates credential theft as an attack vector.
- Personal information (name, year, place) dramatically reduces the effective keyspace in targeted attacks.
- Password reuse is a systemic vulnerability; one breached site can cascade to all other accounts.

10. Conclusion

This task provided a comprehensive, hands-on understanding of password security. Through evaluating six passwords across the strength spectrum, it became clear that password strength is not merely a checklist of character types, but a function of entropy — the mathematical measure of unpredictability. The tools confirmed that common patterns (dictionary words, leet-speak, predictable capitalisation) are explicitly targeted by modern cracking software, rendering many passwords that appear complex to humans trivially crackable.

The key takeaway is that length, randomness, and uniqueness are the three pillars of strong passwords. Passphrases provide an elegant solution for human-memorable high-entropy credentials. Password managers and MFA represent the practical deployment of these principles at scale, and their adoption is the most impactful security upgrade any individual or organisation can make.

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